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PSYCHOLOGY

The nature (and nurture) of IQ

New ideas about intelligence rocked the nascent field of academic psychology in the 1930s

By **Meredith Wadman**

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As infants, the two toddlers under evaluation—13-month-old CD and 17-month-old BD—had been taken by the state from their mothers, a prostitute and an inmate in an insane asylum, respectively, and placed at the Davenport home, where they were "scarcely touched, never held, rarely spoken to," as Marilyn Brookwood, the author of the excellent *The Orphans of Davenport*, writes. When Skeels performed an IQ test, the girls scored 46 and 35. (An individual score of 90 to 109 was considered average intelligence.)



The Orphans of Davenport
Marilyn Brookwood
Liveright, 2021.
352 pp.

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The women at the Woodward State Hospital for Epileptics and School for the Feeble-minded lavished the children with affection. Nine months later, Skeels was astonished to find CD and BD "alert, attractive, playful, [behaving] like any normal toddlers," Marie Skodak, a colleague who accompanied him that day, later wrote. After another 18 months living with the women, CD's IQ score was 95 and BD's was 93. The girls were returned to Davenport and adopted within months.

When located again in their late 20s, both were married with children in apparently stable, loving households.

CD and BD are among the many children whom Skeels and a handful of colleagues at the University of Iowa's Iowa Child Welfare Research Station studied in the 1930s. Benefiting from the station's milieu of intellectual freedom, these scientific heroes of *The Orphans of Davenport* developed a body of work finding that neglected children placed in caring and stimulating environments could recover tens of IQ points; that institutional neglect eroded, but preschool

improved, children's IQ scores; and that institutionalized babies born to low-IQ parents and adopted in the first months of infancy scored in the good or superior range on later IQ tests.

The suggestion that children's IQ scores changed with time and circumstance set off a frenzy of reaction in the young, insecure discipline of academic psychology. Its eugenicist establishment was convinced that intelligence was a fixed and heritable trait—a belief that generated between 60,000 and 70,000 forced sterilizations in the United States in the 20th century. Led by Stanford University psychologist Lewis Terman, the field's leaders launched sustained, vicious, and humiliating attacks on the Iowa group.

It would be 30 years before the validity of the Iowa group's clinical findings and their implications—that nurture as well as nature plays a pivotal role in the development of children's intelligence—were at last recognized and celebrated by their profession. But Skeels and his Iowa colleagues—chief among them Skodak, the outspoken daughter of Hungarian immigrants; Beth Wellman, who conducted pioneering preschool studies before dying prematurely; and the research station's director, George Stoddard, who would go on to become the chancellor of New York University—made a lasting impact all the same. Their work caught the attention of the Kennedy and Johnson administrations, helping to launch Head Start, the US government program for disadvantaged preschoolers.

The quiet courage of the Iowa researchers illuminates this story, not least when others admit that they failed to rise to the same challenge. "I feel guilty" for finding results similar to those of the Iowa group but not publicizing them in the 1930s, recalled leading children's development scholar Lois Barclay Murphy decades later. "I knew that Skeels and Skodak were sound, but with the whole establishment against us I didn't think there was any point in trying to convince people."

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READING

READING PASSAGE 1

You should spend about 20 minutes on Questions 1–13, which are based on Reading Passage 1 below.

Early Childhood Education

New Zealand's National Party spokesman on education, Dr Lockwood Smith, recently visited the US and Britain. Here he reports on the findings of his trip and what they could mean for New Zealand's education policy

A

'Education To Be More' was published last August. It was the report of the New Zealand Government's Early Childhood Care and Education Working Group. The report argued for enhanced equity of access and better funding for childcare and early childhood education institutions. Unquestionably, that's a real need; but since parents don't normally send children to pre-schools until the age of three, are we missing out on the most important years of all?

B

A 13-year study of early childhood development at Harvard University has shown that, by the age of three, most children have the potential to understand about 1000 words – most of the language they will use in ordinary conversation for the rest of their lives.

Furthermore, research has shown that while every child is born with a natural curiosity, it can be suppressed dramatically during the second and third years of life. Researchers claim that the human personality is formed during the first two years of life, and during the first three years children learn the basic skills they will use in all their later learning both at home and at school. Once over the age of three, children continue to expand on existing knowledge of the world.

C

It is generally acknowledged that young people from poorer socio-economic backgrounds tend to

do less well in our education system. That's observed not just in New Zealand, but also in Australia, Britain and America. In an attempt to overcome that educational under-achievement, a nationwide programme called 'Headstart' was launched in the United States in 1965. A lot of money was poured into it. It took children into pre-school institutions at the age of three and was supposed to help the children of poorer families succeed in school.

Despite substantial funding, results have been disappointing. It is thought that there are two explanations for this. First, the programme began too late. Many children who entered it at the age of three were already behind their peers in language and measurable intelligence. Second, the parents were not involved. At the end of each day, 'Headstart' children returned to the same disadvantaged home environment.

D

As a result of the growing research evidence of the importance of the first three years of a child's life and the disappointing results from 'Headstart', a pilot programme was launched in Missouri in the US that focused on parents as the child's first teachers. The 'Missouri' programme was predicated on research showing that working with the family, rather than bypassing the parents, is the most effective way of helping children get off to the best possible start in life. The four-year pilot study included 380 families who were about to have their first child and who

represented a cross-section of socio-economic status, age and family configurations. They included single-parent and two-parent families, families in which both parents worked, and families with either the mother or father at home.

The programme involved trained parent-educators visiting the parents' home and working with the parent, or parents, and the child. Information on child development, and guidance on things to look for and expect as the child grows were provided, plus guidance in fostering the child's intellectual, language, social and motor-skill development. Periodic check-ups of the child's educational and sensory development (hearing and vision) were made to detect possible handicaps that interfere with growth and development. Medical problems were referred to professionals.

Parent-educators made personal visits to homes and monthly group meetings were held with other new parents to share experience and discuss topics of interest. Parent resource centres, located in school buildings, offered learning materials for families and facilitators for child care.

E

At the age of three, the children who had been involved in the 'Missouri' programme were evaluated alongside a cross-section of children selected from the same range of socio-economic backgrounds and family situations, and also a random sample of children that age. The results were phenomenal. By the age of three, the children in the programme were significantly more advanced in language development than their peers, had made greater strides in problem solving and other intellectual skills, and were

further along in social development. In fact, the average child on the programme was performing at the level of the top 15 to 20 per cent of their peers in such things as auditory comprehension, verbal ability and language ability.

Most important of all, the traditional measures of 'risk', such as parents' age and education, or whether they were a single parent, bore little or no relationship to the measures of achievement and language development. Children in the programme performed equally well regardless of socio-economic disadvantages. Child abuse was virtually eliminated. The one factor that was found to affect the child's development was family stress leading to a poor quality of parent-child interaction. That interaction was not necessarily bad in poorer families.

F

These research findings are exciting. There is growing evidence in New Zealand that children from poorer socio-economic backgrounds are arriving at school less well developed and that our school system tends to perpetuate that disadvantage. The initiative outlined above could break that cycle of disadvantage. The concept of working with parents in their homes, or at their place of work, contrasts quite markedly with the report of the Early Childhood Care and Education Working Group. Their focus is on getting children and mothers access to childcare and institutionalised early childhood education. Education from the age of three to five is undoubtedly vital, but without a similar focus on parent education and on the vital importance of the first three years, some evidence indicates that it will not be enough to overcome educational inequity.

The columnist

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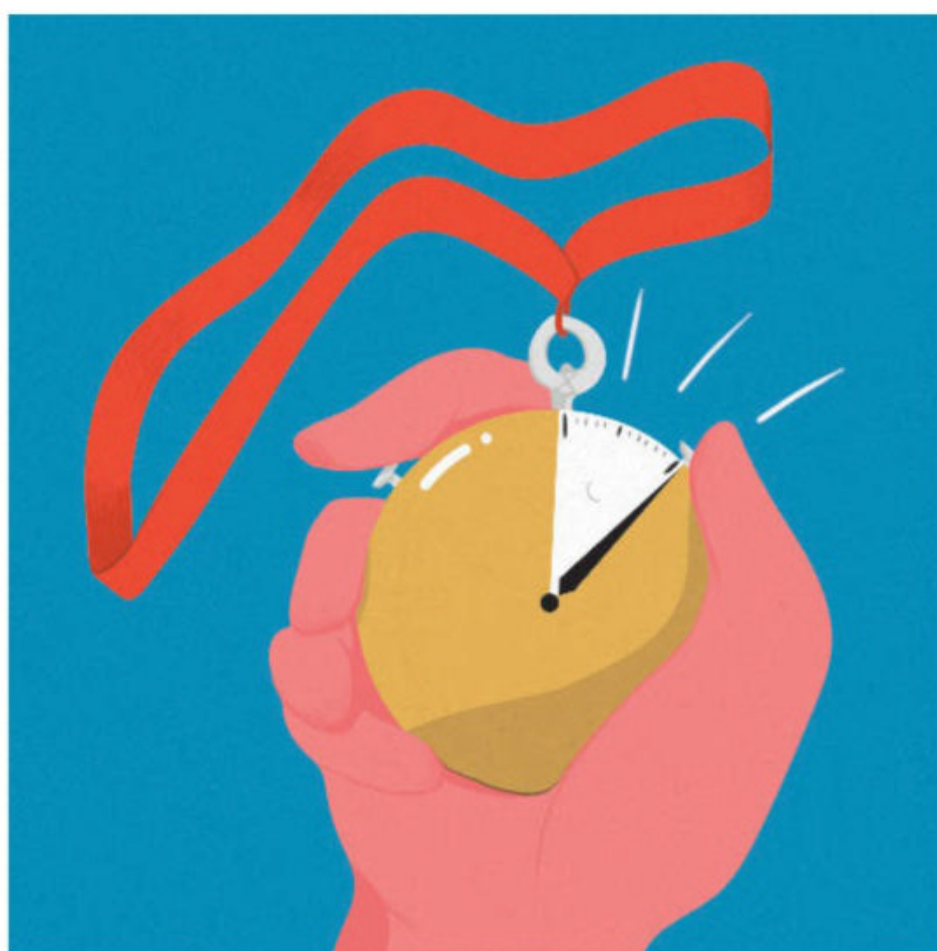
Record-breaking Olympics

World records tumbled at an impressive rate on the athletics track at this year's Olympic games. Was new technology responsible, asks **Laura Healey**

THE Tokyo Olympics brought some of the fastest times ever seen on the athletics track. At an astonishing number of races, athletes beat personal bests along with national, Olympic and world records. Elaine Thompson-Herah set a new Olympic record in the women's 100 metres, breaking Florence Griffith Joyner's record set more than 33 years ago. World records were smashed in both the men's and women's 400 metres hurdles, by Karsten Warholm and Sydney McLaughlin respectively. In both these events, the silver medallist also ran faster than the previous world record.

Was this just an unusually good Olympics for record-breaking races or was something different going on? Part of the answer could be found by looking down at an athlete's feet. If you looked closely, you might have spotted some new technology known as "super spikes" – and underfoot, there was a high-tech track.

Recently, track spikes – shoes that have spikes on the underside to give runners grip – have seen a similar shift in the performance-enhancing technology that previously took place with marathon racing shoes. Marathon "super shoes" first emerged in 2017 with Nike's Vaporfly 4%, which gave athletes average energy savings of 4 per cent compared with competitors not wearing them. By now, almost every brand has a super shoe, and the new technology is



being applied to track spikes.

Similar to their super shoe counterparts, super spikes combine soft, compliant and resilient foam with a stiff, curved carbon-fibre plate. The exact benefits of super spikes are difficult to quantify, but each component probably plays a role.

Traditionally, track spikes have tried to lessen the amount of midsole foam to reduce weight and energy absorption. However, new technology is lightweight and the foam is better at returning energy to the athlete than foams before it, giving back as much as 80 to 90 per cent. In this way,

the foam acts as a spring with each step the athlete takes.

The role of the carbon-fibre plate is less clear. Research has shown that stiffening track spikes will reduce the amount of energy lost during toe flexion. This may increase the demands on the ankle. However, it has also been shown that if an athlete is strong enough to meet these increased demands, the plate allows them to get a more effective push off.

Another new technology that contributed to athletes' speed in Tokyo was the track. While it may have looked like a regular track, Mondo, the company

behind it, spent three years researching and developing the surface specifically for Tokyo 2020. The track was precisely tuned to allow shock absorption and energy return, playing a similar role to the foam in spikes.

Research on engineered track surfaces is as old as 1978, when it was found that a compliant, resilient track surface could improve times by as much as 2.9 per cent. Although the exact savings of the Tokyo track are unknown, its developers have said it could improve times by as much as 2 per cent compared with previous Olympic tracks.

While it is tempting to attribute the record-breaking times to the new spike and track technologies alone, other factors were at play too. The games being delayed by a year due to the covid-19 pandemic may actually have benefited some athletes, giving them more time to train. Other, more variable, factors, such as the weather, can affect how an athlete performs on any given day. And above all, we cannot ignore the effort and talent of the athletes competing in these races. It may just be that we got a particularly talented crop this year.

Overall, the record-breaking times seen in Tokyo are likely to be a combination of all the above, including fast shoes, fast tracks and extremely talented athletes. ■



Laura Healey is a manager of footwear innovation at Puma

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It is probably best
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Leapfrogging fossil fuels

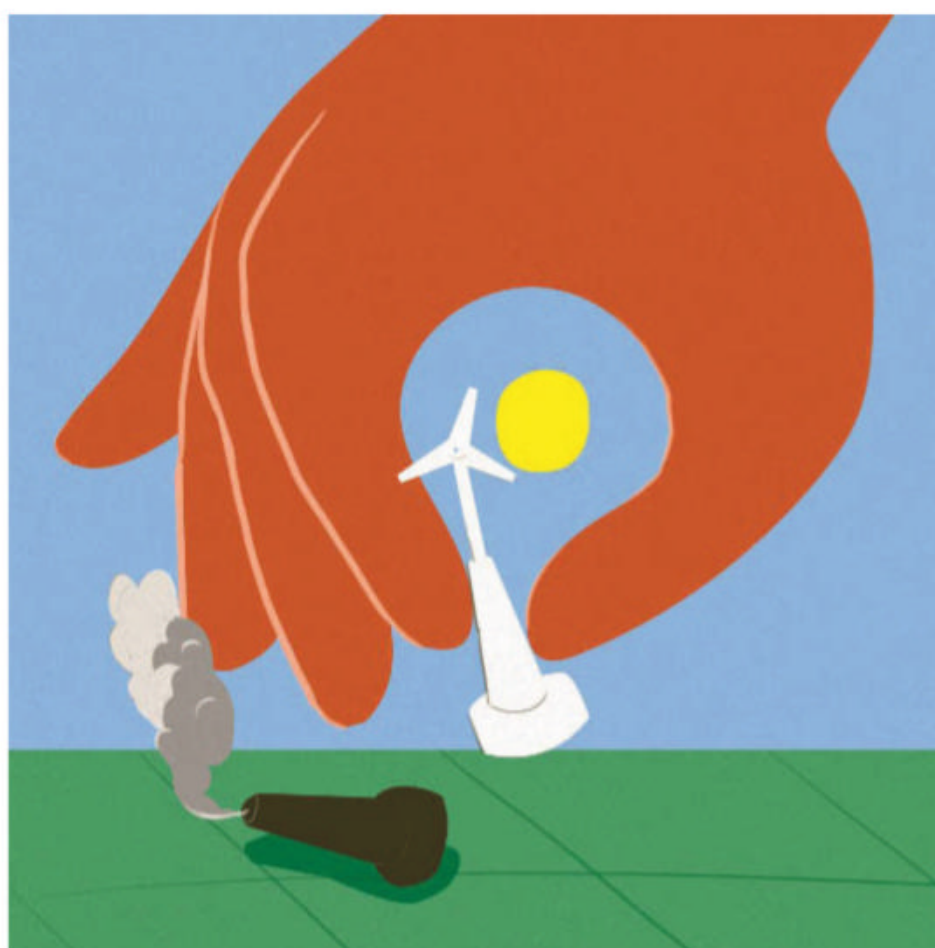
As energy access improves in low-income countries, there is an opportunity to go straight to clean technologies, says **Jim Watson**

THE world has an energy dilemma. On the one hand, we need to drastically clean up energy use in higher-income countries to tackle climate change. But on the other, there are still millions of people who don't have reliable access to energy. As their energy access improves, there is a risk that this could offset some of the world's shift to low-carbon energy. It doesn't have to be that way: this is also an opportunity for some countries to skip much of the fossil fuel stage altogether.

For low-income countries, making big improvements in access to electricity is crucial. Better access to energy is linked to improvements in education, economic development and health, for example. According to the latest data from international organisation Sustainable Energy for All, more than 750 million people lack access to electricity and over 2.5 billion people don't have access to clean cooking technologies or fuels. Many more have limited or unreliable access to electricity.

Improving this situation could be a chance to do things differently. Instead of developing energy infrastructures based on fossil fuels, low-income countries could leapfrog straight to cleaner, low-carbon technologies.

This isn't a pipe dream. In the telecommunications sector, for example, landlines never fully took hold in many low-income countries. Instead, people moved



straight to using mobile phones. This has also enabled services such as banking via "mobile money".

The potential for leapfrogging in the electricity sector has been strengthened recently by the steep fall in the costs of renewable technologies, and reductions in the costs of complementary technologies such as batteries. According to a recent report by the International Renewable Energy Agency, the cost of large-scale solar has fallen by 85 per cent in the past decade, while wind power costs have fallen by about 50 per cent. The conventional assumption that fossil fuel

electricity is cheaper is now on its way out, as is the idea that improved access is all about improved access is all about centralised electricity grids.

A good example of this is the solar homes programme in Bangladesh, which extended electricity access to 4 million homes (about 12 per cent of the population) by 2016. This provided much quicker access to services such as electric lighting than extending the centralised electricity grid would have done.

Similarly, off-grid solar systems now provide about 7 per cent of Rwanda's population with access to electricity. Many of these systems

are large enough to power TVs and appliances as well as lighting.

Of course, initiatives like these don't completely solve the energy access problem. To provide the necessary power for industries and all domestic uses, renewables must be deployed at a much larger scale. This requires investment in grid infrastructure, too. It will be important to avoid repeating the recent experience of Vietnam, where solar and wind investment increased rapidly, but many plants couldn't operate fully because of grid constraints.

So what else needs to happen to make leapfrogging possible? It is crucial to recognise that most low-income countries have very low emissions and their priority has to be expanding energy systems to underpin economic development and universal energy access. Fossil fuels shouldn't be entirely ruled out, but should be part of a wider strategy that prioritises low-carbon investments. That often means reforms to policies and regulations so that low-carbon options aren't disadvantaged, and a big scale-up of finance from donors to reduce the financial risks of adoption. Only then will low-income countries be able to leapfrog to cleaner energy. ■

For more on the new energy world, see our special on page 34



Jim Watson is research director at the UCL Institute for Sustainable Resources in London

The Tokyo Olympics

That 2021 feeling

TOKYO

The games are not a flop. Nor are they a success

JAPAN HAS set several records in recent weeks. Its Olympic team has racked up 21 gold medals, the most in its history. It has a new youngest-ever gold medallist, a 13-year-old skateboarder, Nishiya Momiji. And it has achieved new heights in its daily covid-19 caseload, now exceeding 14,000.

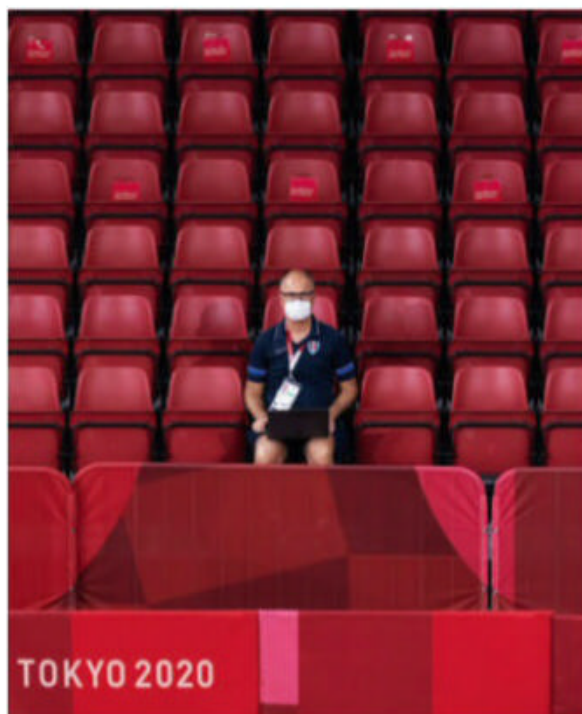
The combination has generated mixed feelings about the Olympics. Many Japanese have celebrated the athletic triumphs, despite misgivings about the games before they started. Barred from attending most events in person, sports enthusiasts have gathered at outdoor venues to try to catch glimpses of cyclists or skateboarders. Television viewing figures have also been strong. But enthusiasm for sports has not rubbed off on the event as a whole or on its organisers, as Japanese leaders had hoped. The approval ratings for Suga Yoshihide, the prime minister, and his cabinet have dipped below 35%—another new record.

These games are likely to be remembered not as a symbol of how humankind overcame the pandemic, but as an example of how strange life became because of it. The oddity has been on display from the start. On opening night, a long column of protesters marched through central Tokyo to the new Olympic stadium. During quiet moments in the ceremony, their chants—“Stop the Olympics”—could be heard inside the mostly empty venue. Farther up the same street, fans gathered, streaming the ceremony on their smartphones and glancing up to catch fireworks and drone shows rising above the stadium’s rafters.

With the events under way, Olympians have shuttled between hotels and venues, where they compete in front of scatterings of volunteers, officials and journalists. Around the host cities, reminders of the games are muted. Absent are the big crowds and good cheer that usually accompany such events.

“It’s a little lonely,” laments Iizuka Masaki, a collector of sports pins waiting outside the heavily guarded gates to the Tokyo stadium in the hope of finding partners from abroad to trade with. When the Japanese and American baseball teams faced off on August 3rd in Yokohama, the loudest noises near the stadium came from the buzzing of cicadas.

Mr Suga has alternated between calling medal-winners to congratulate them and calling press conferences to announce new covid countermeasures. With the Delta



And the crowd goes wild

variant dominant and hospitals under strain, the government decreed this week that only critically ill or high-risk patients would be admitted to hospitals. Others must recover at home. (Deaths have remained relatively low, thanks in part to high vaccination rates among the elderly.) The state of emergency in Tokyo and Okinawa was expanded on August 2nd to include four more prefectures.

Officials insist that the rising case numbers in Japan have nothing to do with the games. Strictly speaking, that may be true. Some 500,000 covid tests have been conducted on Olympic participants, of which just 0.02% have come back positive. Although a few athletes—and the entire Greek artistic swimming team (yes, that is an Olympic sport)—have had to drop out because of positive tests, the competitions have unfolded smoothly. A handful of athletes and support staff have had their credentials revoked for breaking protocol and wandering around, but as yet no evidence has emerged linking reckless Olympians to viral clusters among the wider population.

Since the start of the pandemic, however, Japan’s ability to contain it has relied on co-operation from the public. The government lacks the legal authority to impose strict lockdowns; its emergency declarations are, in essence, requests for people to limit their movements and for businesses to close early and stop serving alcohol. The current state of emergency is the fourth. Fatigue is setting in.

Holding the Olympics has reinforced the complacency. Foot traffic, for instance, has fallen far less during the current state of emergency than it did during past ones. “Since the Olympics are being held, it makes people think it’s okay to go out,” says Onishi Moeri, who joined a crowd taking pictures beside the Olympic rings early this week. It has been, muses Mr Iizuka, “a very unusual Olympics”. ■

Covid-19 in Myanmar

The second horseman

SINGAPORE

The Burmese army is making a bad pandemic worse

THAN THAN SOE has not had a day off since covid-19 began charging through Myanmar at the end of May. She runs a charity in Yangon, Myanmar’s biggest city, which transports covid patients to hospitals and corpses to cemeteries. Her organisation collects as many as 60 bodies a day. “We are very tired but we keep going,” says Ms Than Than Soe. Two volunteers recently got infected and died.

Covid is ravaging Myanmar. Daily cases, taken as an average over seven days, have hovered around 5,000 since mid-July, but limited testing means that this is probably an underestimate. The share of tests that return positive results has exceeded 35% since mid-July, which suggests widespread, uncontrolled transmission. Only Iran and Mexico have higher positivity rates. Myanmar, which borders countries that are home to about a third of the world’s population, risks becoming a “super-spreader state”, according to Tom Andrews, the UN’s expert on the country.

Almost every South-East Asian country is experiencing its worst outbreak yet, and for many of the same reasons: low rates of testing and vaccination, weak or overwhelmed health-care systems and the presence of more transmissible variants such as Delta. But Myanmar offers the virus particularly hospitable conditions, thanks to the army. After a coup in February, testing, contact-tracing and treatment of covid ground to a halt. Public hospitals emptied of medical workers, thousands of whom joined protests against the coup. When a state bordering India reported an outbreak in June, the junta was too busy quelling resistance to do much about it.

In any case, lockdowns are impossible to enforce when people believe their lives are in danger. Violence unleashed by the coup has spurred many Burmese to take flight. Some 230,000 people fled their homes between February and June, bringing the total number of displaced people in Myanmar to 680,000. The camps where some have found refuge often have limited health care. The jungles where others flee have none.

But the struggle to find medical assistance has become a national problem, shared by refugees and city folk alike. Myanmar had few doctors to begin with. In 2018 there were just 0.7 for every 1,000 people—fewer even than in India, which had 0.9. Their number has been depleted by the ▶

China's coronavirus vaccines

Three key Chinese vaccines are in use outside of China, particularly in low-income countries, but their efficacy varies

IN TERMS of sheer numbers, the leading Chinese vaccine is the snappily named BBIBP-CorV from Beijing-based pharmaceutical firm Sinopharm and the Beijing Institute of Biological Products. This is an inactivated virus vaccine that uses viruses originally isolated from covid-19 patients in China. These are grown in cell culture and then “killed” by soaking them in a chemical called beta-propiolactone.

This vaccine is approved in some form in 59 countries, mostly low-income ones, and also has emergency use listing from the World Health Organization (WHO), which means it can be used under COVAX, the global initiative for equitable vaccine sharing.

Information from Sinopharm is thin on the ground, but the WHO says that all the vaccines on the emergency use list are “highly effective in preventing severe disease and hospitalisation due to covid-19”. WHO approval is also an endorsement of safety.

Peer-reviewed results of a phase III trial showed that the vaccine is 78 per cent effective against symptomatic disease (*JAMA*, doi.org/gj7khd). The WHO concluded that the Sinopharm coronavirus vaccine is 79 per cent effective against hospitalisation.

The vaccine seems to be effective against the alpha coronavirus variant but less so against the beta variant. Peer-reviewed data on its effect on the gamma and delta variants has yet to emerge. The WHO says there is also no information available on whether it prevents transmission.

Another inactivated virus vaccine from China, Coronavac from Sinovac Biotech, has some form of approval in 39 countries, mostly low-income ones, and has WHO emergency use listing.

Efficacy of this vaccine varies a lot from trial to trial, with a high of 84 per cent against symptomatic disease in a trial in Turkey and a low of just over 50 per cent in Brazil. The WHO's emergency use listing assessment concluded that the vaccine is 51 per cent effective at preventing symptomatic disease but 100 per cent effective against severe disease and hospitalisation.

Effectiveness against variants also varies. Some research suggests the Sinovac vaccine works well against the alpha variant but less so against beta, and is ineffective against gamma. However, the WHO says it is effective against gamma. There is no data on

Children aged 12 to 17 queuing for vaccination in Suqian, China, in July

its efficacy against delta. The WHO also says there is no information available on whether it prevents transmission.

A third Chinese vaccine, Ad5-nCoV from CanSino Biologics, is approved in eight countries but not by the WHO. It uses an adenovirus as a parcel to deliver the DNA for the coronavirus spike protein into the body, given as a single dose. Results

“All the vaccines on the WHO emergency use list are effective in preventing severe disease”

from a trial in Pakistan suggest 66 per cent effectiveness.

Three other Chinese vaccines are being used exclusively in China, and one is in use in China and Uzbekistan. One of these vaccines was also developed by Sinopharm, in partnership with the Wuhan Institute of Virology. It is essentially identical to BBIBP-CorV and just as effective, but used a different SARS-CoV-2 sample as its starting point.

Graham Lawton

More in the pipeline

Several further coronavirus vaccines are in late-stage testing. According to McGill University's vaccine tracker, there are 130 coronavirus vaccines still in clinical trials, 40 of them in phase III.

In the UK, those most likely to see the light of day are made by Novavax, Valneva and CureVac. The UK government has options to buy batches of all three, pending approval.

Novavax's shot consists of spike proteins from the SARS-CoV-2 virus embedded in a virus-like nanoparticle. Interim results from phase III trials suggest it is about 90 per cent effective at preventing disease with the original coronavirus strain but less effective with variants (*NEJM*, doi.org/gk3zvz). Novavax is tweaking the vaccine to deal with them.

Valneva's vaccine is an inactivated virus vaccine, the only one of its kind in development in Europe. It is in a phase III trial in the UK, with results expected in September.

CureVac's jab is an mRNA vaccine, but results from its phase III trials so far have been disappointing, with only 48 per cent efficacy against infection.

Other late-stage vaccines worth watching are from the Canadian firm Medicago, which grows vaccines in a relative of the tobacco plant, as well as a viral protein vaccine from pharma giants Sanofi and GlaxoSmithKline. Initial results on the latter disappointed and trials were halted, but it has been reformulated and is showing promise in phase III trials. **Graham Lawton**



COSTA/PHOTO/BARCROFT MEDIA VIA GETTY IMAGES

◀ from paid product placement, and consumer behavior continues to pivot toward media consumption in ad-free/ad-light environments, I would anticipate more investment will follow,” James Clarke, a senior director of media strategy at Frito-Lay North America, said in an email.

Still, more data-driven hypertargeting of product placement could stoke the same kind of consumer pushback that online digital ads have received. CEO Butler is aware the practice could cause some discomfort, especially when it’s bankrolled by Gates, one of the 10 richest people on Earth. But

he says that the complex data his company uses to help marketers is all about explaining how groups—rather than individuals—react, while conceding that his focus is on ensuring his company’s product placements turn into sales.

“It’s the next frontier of advertising,” Butler says. “Because of all this data, you can get way more impact and guarantees that people are actually seeing the content you’re in.” —*Kelly Gilblom*

THE BOTTOM LINE Product placement spending is projected to reach \$23.3 billion this year. It could grow even more as marketers look for ways to reach consumers who pay for ad-free streaming.



● Butler

Grounding China’s Personal Shoppers

● A government shift and Covid travel restrictions have stalled *daigou* who buy goods overseas for Chinese consumers

Vivian Li, a stay-at-home mother in Shenzhen, used to make weekly trips abroad, leaving her husband in charge of their toddler and traveling to Hong Kong or Tokyo with two empty suitcases. She would fill them with Lancôme eye creams, Pola shampoos, Louis Vuitton handbags, and other products that in China were either much pricier or simply unavailable—sometimes clearing out the store shelves for items that internet celebrities had recommended. Then she would return home and send the goods via express delivery to clients who’d hired her to shop for them. Working as a personal shopper, or *daigou*, was a lucrative job for Li, 28, who could make more than 30,000 yuan (\$4,626) a month from her travels.

This army of gray-market surrogate shoppers has long been a feature of China’s retail sector, serving consumers who crave items that aren’t available locally or are sometimes significantly more expensive in the country. Despite the government’s best efforts to kill off the *daigou* trade through stricter customs checks and eased taxation rules for legitimate cross-border e-commerce channels, it grew to \$40 billion in revenue in 2019, according to the consultancy Proresearch in Beijing.

Then came Covid-19. China slammed its borders shut to keep out the virus, putting a halt to most overseas travel—and the business of Li and other *daigou* operators. Although she places some orders through overseas partners, without Li on-site to personally vouch for the provenance of the products, many clients don’t trust the authenticity of the goods. That skepticism has led her to lose more than half her business. “Many of my *daigou* peers

have exited the market,” she says. “I just hang in there hoping the border can reopen soon, but the longer we wait, the more likely my clients have found other channels and leave forever.”

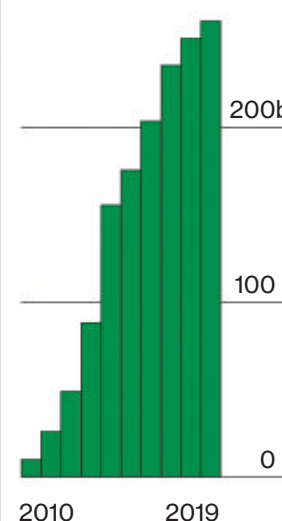
The collapse of *daigou* has also left businesses, whether infant formula makers or cosmetics retailers, struggling. Before the pandemic, some foreign companies grew rapidly thanks to the trade. By selling to personal shoppers and letting them handle the delivery back to China, they didn’t have to invest in building local retail operations of their own. Companies could also avoid requirements to comply with Chinese labeling and packaging laws.

“The whole *daigou* sector has dried up,” says Sage Brennan, co-founder of China Luxury Advisors, a consulting company in Los Angeles. “A lot of consumers have gotten into more legitimate cross-border e-commerce, and that becomes easier as time goes on than relying on the gray markets.”

Companies that counted on *daigou* sales are now saddled with excess inventory. “*Daigou* is unlikely to reach its heyday market size anymore even after the pandemic eases,” says Ethan Ye, a principal in Shanghai for consultants Oliver Wyman. One reason: Luxury brands that focused on expanding in China during the pandemic are putting more pressure on *daigou* rivals, says Max Peiro, chief executive officer of Re-Hub, a data analytics company in Shanghai. “More and more brands are taking active measures” against the gray-market sellers, he says. “They realize this is a situation they need to try to mitigate.”

One of the most popular *daigou* products is infant formula, and during the offshore shopping

▼ Size of China’s overseas *daigou* shopping market, in yuan



industry's peak time, from 2010 to 2019, Australia, Hong Kong, and other places restricted the number of containers individuals could purchase to prevent daigou shoppers from buying up all the supply. But A2 Milk Co. in New Zealand downgraded its forecast in May for the second time in five months, after its revenue in Australia and New Zealand fell about 30% in the second half of 2020 because of disruption to daigou channels. The share price of the company, among the largest on New Zealand's stock market, has fallen more than 60% over the past 12 months, compared with an 8% rise for the benchmark index.

The outlook for A2 and rivals Danone SA and Reckitt Benckiser Group Plc is grim as the delta variant spreads and travel curbs persist. "The longer this happens, the harder it will be for Danone, Reckitt and A2 Milk to rebuild China sales as their local customer bases shrink," Bloomberg Intelligence analysts Catherine Lim and Kevin Kim wrote in a report published on July 5.

Even before the pandemic, there were signs the business faced tougher times. Worried about lost revenue, China's government tightened customs checks and tax regulations. Chinese consumers also were gaining more options domestically, a result of the proliferation of cross-border e-commerce services and duty-free malls.

Some customers might be lost for good as travel curbs change consumer behavior. About 60% of Chinese consumers last year used cross-border e-commerce platforms, led by Tmall International and JD International, according to IResearch. Those online retailers had import sales of more than 200 billion yuan in 2020, and their sales should grow about 30% annually over the next two years, even as the pandemic eases, IResearch says.

Japanese drugstore operator Tsuruha Holdings Inc., popular among daigou customers for its beauty and personal-care products, is turning to online platforms. In September it opened a store on WeChat, the app that Tencent Holdings Ltd. owns, to deliver products directly to customers in China. At the top of every product page of its WeChat store are the Chinese words "Shipped from Overseas, Authenticity Guaranteed," because even the most price-sensitive daigou consumers worry about fake products and shoddy quality.

To discourage customers from going back to daigou, some brands are offering more services designed to build longer-term relationships with individual consumers. Japanese cosmetics maker Shiseido Co. has appointed beauty consultants to provide one-on-one services to its Chinese VIP members and to send them free product samples.

Shanghai-based Ushopal, founded in 2017, works with more than 3,000 online influencers and helps organize popup stores to promote items such as perfume from Juliette Has a Gun and lipstick by Suqqu. Customers placing orders via Ushopal's cross-border e-commerce platform can get delivery in four hours from its Shanghai-area warehouse.

For luxury brands in Europe, daigou hindered efforts to build relationships with customers; the buyers purchased in bulk for clients, so the brands couldn't identify them. "It's all about knowing the local client now and establishing a long-term relationship," says Jonathan Siboni, CEO of data intelligence firm Luxurysight in Paris. "It's more than just a one-off handbag sale."

Some brands have raised prices in Europe to narrow the advantage over the prices of their goods in China, undercutting one of the main reasons the



daigou started in the first place. "You don't buy with daigou when you only have a price differential of 15% to 20%," Siboni says. "There was a rationale if the differential was 50%, but that's no longer the case."

Still, Li and other daigou say they expect at least some customers will return. A Japanese beauty retailer Li works with is getting ready, opening several warehouses in China for some hot-selling products to shorten the delivery time to consumers. Li's job will be to drum up business through her social network. It won't be like the old days of overseas shopping, she says, but that's all right. "Most customers are fine to pay" the import duties, Li says. "They still can't find some of the products in domestic channels anyway." —Bruce Einhorn, Daniela Wei, and Angelina Rascouet

THE BOTTOM LINE Daigou shoppers, who make purchases offshore for customers in China, were a \$40 billion business. The pandemic travel halt is fueling the practice's decline.

How Chinese Retailers Are Reinventing the Customer Journey

Five lessons for Western companies

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TECHNOLOGY





TECHNOLOGY

China is both a large and a fast-growing retail market—worth about \$5 trillion in 2020—and highly digitized.

Given that the pandemic has made digital every retailer's strategic priority, it's not hard to see why the *Economist* opened 2021 with a cover story headlined "Why Retailers Everywhere Should Look to China."

In China online sales have grown about 25% in each of the past seven years and reached about \$1.9 trillion in 2020, when they amounted to some 25% to 50% of total retail (compared with 10% to 20% in the United States). More than 90% of those sales are on mobile devices, compared with less than 50% in the United States. So it should be no surprise that Chinese companies and individuals have led the way in developing video retail, social commerce, community retail, retail-as-a-service, and many other new digital channels, including the super app, which provides an all-in-one experience for consumers by accessing various services and offerings.

Who are these leaders?

There's Douyin—known in the West as TikTok—which started as an entertainment app for sharing short videos and soon discovered that many users were commenting on popular videos by creating their own versions. Douyin encouraged participation by welcoming content creators, who often featured their favorite products and clothing styles, making the app a marketing tool. China's livestreaming market has

reached \$16.3 billion and is now integral to how people shop—which is why Walmart invested in Douyin.

Pinduoduo, the largest agriculture-focused platform in China, was founded in 2015 and is currently worth \$175 billion. Sometimes described as "Groupon on steroids," it has gamified the shopping process, enabling groups to haggle with merchants, often via WeChat.

Then there's Li Jiaqi, a 28-year-old influencer who pioneered digital cosmetics retailing and is known as the Lipstick King. He boasts more than 7 million followers on Weibo and close to 40 million on Douyin. He once sold 15,000 lipsticks in just five minutes and tried on 380 lipsticks in a seven-hour livestreaming show. Li Jiaqi is the only male key opinion leader (KOL) for cosmetics and the best salesperson for beauty products in China. A former beauty adviser at L'Oréal, he has phenomenal influence: His recommendations can make or break a product launch. He demonstrates and recommends the products of multiple companies and is paid according to the sales he generates.

These innovators owe much of their success to the massive ecosystems of Alibaba, JD.com, Tencent, and, increasingly, Pinduoduo, ByteDance, and Meituan, which serve as key touchpoints for consumers. They attract Chinese retailers and international brands by leveraging ever more data in innovations such as Alibaba's Tmall Smart Selection (a product recommendation algorithm) and Meituan's highly sophisticated logistics routing algorithm. As Tencent's founder, Pony Ma, put it in an internal presentation in early 2020, "The era of smart retailing ignited by 'super connectivity' just kicked off." (That said, China's government is taking a harder line with its biggest online enterprises. In late 2020 it blocked the IPO of Ant Group, a fintech company spun out of Alibaba, at the last minute. Alibaba was also fined some \$2.8 billion in April 2021 under the PRC's antitrust law.)

Retailers in Europe and the United States don't have access to such integrated ecosystems, but they can still usefully borrow from China's innovators. To identify how, we studied 25 Chinese digital retail companies, including the giants Alibaba, JD.com, Meituan, and Tencent; the emerging platforms ByteDance, Pinduoduo, and Ele.me; and successful brands such as Peacebird, Forest Cabin, Babytree, Soufeiya, and Xtep. In addition, our research benefited greatly from discussions with dozens of executives from



In China online sales have grown about 25% in each of the past seven years and reached about \$1.9 trillion in 2020, when they amounted to some 25% to 50% of total retail.

Western companies, including Amazon, Apple, Daimler, Luxottica, Nestlé, Nike, PayPal, Philips, Siemens, Starbucks, and Walmart, along with Japan's Sogo & Seibu (department stores) and Russia's X5 Retail Group (a leading food retailer).

In this article we draw on that research to explain five lessons that Western companies can learn from China as they develop their own digital market offerings.

1 CREATE SINGLE ENTRY POINTS

A single point online where customers can access all their potential purchases is the holy grail for retailers. China's digital giants have come close to achieving it by creating commerce ecosystems, general platforms offering portals for independent brands, and proactive automated product recommendations.

Ecosystems. For most digital retail consumers in China the first port of call is Taobao (Alibaba's mobile C2C portal) or Alipay, both of which give access to Alibaba's full ecosystem. Alipay, which is on almost every smartphone in China, integrates the platforms and service offerings of companies in Alibaba's huge retail network, enabling consumers to pay for any product or service they may find there, from Nike shoes to wealth management. Tencent's WeChat Pay provides similar benefits. Although users may not go online intending to make a transaction, WeChat Pay's deep integration with external platforms and specific brands means that they often end up doing just that. WeChat has pioneered a concept that allows any brand to develop dedicated but simple subapplications within the WeChat ecosystem. These typically fulfill functions such as e-commerce, coupons, and

task management and are searchable on WeChat's home page. After the initial success of this innovation, most other super apps adopted it.

General platforms. In China's B2C e-commerce market, the world's largest, most independent retailers access customers through platforms provided by Tmall, which accounted for 64% of China's total B2C e-commerce market in 2020, and by JD.com (26%). But players such as VIP.com, Xiaohongshu (RED), and Ymatou are increasingly winning share with specialized platforms: Ymatou focuses on cross-border commerce in high-quality branded products, and RED, often dubbed a lifestyle sharing platform, targets Generation Z and combines social media and e-commerce. China's increasingly sophisticated digital consumers go to these sites for information and to access a community of like-minded customers.

Proactive product recommendations. Digital has shifted power relationships in retail. Recommendation algorithms, livestreaming by KOLs, and "native" (embedded online) e-commerce stores in TikTok have changed the traditional dynamic whereby a purchase begins with the customer's search for a product. For instance, Tmall Smart Selection uses an AI-powered algorithm backed by deep learning and natural language processing to recommend products to shoppers; it then communicates consumer interest to retailers so that they can increase inventory to keep up with demand. That kind of functionality reflects the growing power of retailers relative to manufacturers.

Chinese retailers go further than Amazon does in aggressively leveraging partners and third-party providers (including for logistics, which Amazon mostly manages

IDEA IN BRIEF

THE CHALLENGE

Retailers in Europe and the United States are far behind China in digital retail.

WHY IT EXISTS

China's large and fast-growing market not only is highly digitized but also takes advantage of the massive ecosystems of Alibaba, Tencent, and others.

THE SOLUTION

To catch up, Western retailers can take five lessons from the Chinese experience: Create single entry points, embed digital evaluation in the customer journey, don't think of sales as isolated events, rethink the logistical fundamentals, and always stay close to the customer.



itself), rather than building in-house capabilities. The result is that Chinese retailers are already profitably offering at scale most of Amazon's newer, not yet scaled-up offerings, such as health care, insurance, online groceries, smart home devices, and fashion.

Options for the West. The main challenge for established Western retailers is to get closer to where the consumer is online rather than direct digital traffic to where products can be purchased. To be sure, this is simpler in China because of its less-stringent data privacy rules. But Western companies could do more with the data available to them—for example, by using blockchain to guarantee privacy.

Companies in Europe, given their experience with the EU's tight General Data Protection Regulation, are well-placed to build an early advantage. The Otto Group, in Germany, is a case in point. A catalog retailer for decades, it initially struggled with the arrival in 2008 of Amazon and Zalando, a German online shoe and fashion company. But Otto made early moves into digital solutions, in particular leveraging ventures such as Risk Ident (fraud prevention)

and Picalike (visual search) to build a platform with thousands of partners. Its online revenue for the fiscal year ending in February 2021 reached \$17.8 billion.

Another example is Lydia, an emerging financial-services super app from France, which offers peer-to-peer payment, flexible subaccounts, virtual cards for Apple Pay and Google Pay, and many other functions typically associated with Chinese super apps. To be sure, Europe and the United States have always had well-developed credit card services. But the real potential of super apps is not that they can replace credit cards or cash; it is that they provide a single entry point for consumers, incorporating financial services (lending, investment, insurance); e-commerce; goods delivery and tracking; ticketing for movies, live shows, airplanes, and trains; health care services (hospital reservations, medical consultations, pharmacies); taxi hailing; bike sharing; and a wide range of government functions, including taxation.

The same potential exists in Europe and the United States. Beyond privacy, we see few economic, political, or social barriers to one-stop solutions in the West. The cost to



ABOUT THE ART

Inspired by the booming popularity of online sales, Huang Qingjun asked people from across China to assemble everything they've ever bought online.



TECHNOLOGY

establish comprehensive platforms is relatively low, so new entrants will seize the opportunity if incumbents don't.

2 EMBED DIGITAL EVALUATION IN THE CUSTOMER JOURNEY

A key challenge for retailers is ensuring that consumers can efficiently and effectively evaluate their products in a transparent and unbiased way. With its strong emphasis on influencers and social media, Chinese retail evaluation is highly sophisticated and provides content much richer than what is available in the West.

Customer reviews. Simple scoring or comments on TripAdvisor and Amazon are no longer the standard for Chinese consumers, who provide thousands of detailed comments about products, brands, and shops on JD.com, Tmall, and Taobao, often with photos or videos attached.

Influencers. In the United States people follow key influencers such as Taylor Swift and Kylie Jenner primarily on Instagram, whereas China's opinion leaders have a presence on Weibo, WeChat, TikTok, and other platforms. A common criticism directed at Chinese influencers is that many of them are sponsored—but their neutrality isn't highly valued in Chinese culture. The relationship of an influencer and a consumer is that of a celebrity and a fan, and consumers often consult several influencers before making a decision.

Video livestreaming. Chinese consumers also frequently consult livestreamers, who consist of three main types: CEOs such as Dong Mingzhu of the appliance manufacturer Gree Electric and Li Bin of the electric car company NIO; movie actors and musical artists such as Li Xiaolu and Liu Tao; and professional sellers such as Li Jiaqi, who sold products worth \$145 million on Singles' Day in 2019, and Viya, known as the number one "sister" of Taobao Live.

Independent platforms. In reaching a purchasing decision, Chinese consumers rely far more on independent knowledge platforms such as Zhihu and Zhishi Xingqiu (which are similar to Wikipedia and Quora) than do Western consumers, who might consult *Consumer Reports* for high-value products.

Options for the West. As Chinese consumers have become more-sophisticated users of tech products, they've started consulting multiple data sources. Western companies

need to establish a presence on all the channels where consumers evaluate products—particularly video-based social media. That way they can overcome mistrust of promotional messaging and positive reviews on company sites. Currently the most popular livestreaming in the West runs on gaming platforms such as Twitch, whose tie-in with Amazon makes it a platform for influencer sales. Meanwhile, NTWRK, a mobile-first video shopping platform, features limited-edition products of brands such as Adidas, Guess, and Vans. On its platform the video game company Rooster Teeth and other popular influencers drive sales through product collaborations. Other Western retailers might consider this approach.

3 DON'T THINK OF SALES AS ISOLATED EVENTS

Providing a seamless experience when and where the consumer chooses can radically increase the chances of purchase. In the digital realm China has achieved this in three ways.

Deeply integrated online and off-line sales channels. At Alibaba's Hema Fresh supermarket, for example, consumers can make purchases while sitting at home, on the way to the market, or in the store. They might have fresh food delivered or decide to pick it up while shopping for other products. After JD.com invested in the supermarket chain Yonghui, in 2015, it connected Yonghui's off-line supermarkets to the JD Daojia ("JD to Home") application, which was launched that year in collaboration with Dada, a local on-demand delivery company. (The two merged the following year.) This level of integration brings big advantages. For instance, when the billion-dollar fashion retailer Peacebird closed down its brick-and-mortar stores during the pandemic, it could quickly pivot to online selling, because its sales system no longer differentiated between online and off-line.

Continuous purchasing opportunities. In China a purchase can be made at almost any point in an individual's entire online experience. A consumer might buy directly from an official WeChat account while chatting with friends or in one of WeChat's "mini program malls" or Alipay's mini program for a brand that has advertised in a friend's circle or been recommended in an alumni group. Entertainment and



TECHNOLOGY



A same-city order with a retail chain takes less than half a day to arrive in a large urban center like Shanghai.

shopping are fully integrated as well. A Chinese consumer watching a TikTok video can click on clothes she likes and end up in a native store. Or she might follow a friend's recommendation on WeChat to buy a product at a discount via Pinduoduo.

AI-enabled interfaces. AI-powered chatbots such as Dianxiaomi, which can understand more than 90% of customers' queries, are widely used in China; they did most of the talking during its "Singles' Day" on November 11, 2020, when Alibaba's online transactions exceeded \$74 billion. After-sales service, in terms of delivery, returns, and warranty, was conducted primarily online. According to recent statistics, 94% of online service at Alibaba is AI-enabled, and it earns customer satisfaction ratings 3% higher than service delivered by staffers.

Options for the West. Legacy businesses that are embracing digital approaches need to cut across the silos of online and off-line selling that have been traditional in the West. Even General Motors, the granddaddy of off-line manufacturing, now has more than 100 social media channels. Walmart, the archetypal bricks-and-mortar retailer, announced in February 2020 that it would combine its store and online buying teams in one omnichannel merchandising group. As Zeina Belouizdad, Google's omnichannel product lead for Europe, the Middle East, and Africa, said in late 2018, "We are living in an omnichannel world. Nothing is linear anymore." Consumers don't distinguish between off-line and online, and it is only for legacy reasons that the distinction still exists. Executives outside China should therefore consider aligning and centralizing their sales systems as part of digital transformation efforts.

The United States and large parts of Europe have digital platforms and promising new ventures that could facilitate such integration. Global retailers could potentially enhance their existing digital solutions by partnering with those businesses. For example, in the Netherlands, one of the largest European e-commerce markets, Jumbo supermarkets and the seafood retailer Schmidt Zeevis are taking advantage of the online supermarket Picnic. Consumers can choose where to buy their products and when, with smooth home delivery or in-store pickup options.

Although France's super app Lydia and other new ventures are promising, companies need to do much more to

broaden purchase options. In China, QR code payment and facial recognition scans are now common on Alipay, WeChat Pay, and many other payment apps. In Europe, however, most retailers use contactless card payment or require payments to go directly through a bank app.

4 RETHINK THE LOGISTICAL FUNDAMENTALS

China combines old-fashioned methods and high-tech software to deliver faster and at lower cost than almost any Western retailer can. A same-city order with a retail chain takes less than half a day to arrive in a large urban center like Shanghai, while local supermarkets can usually deliver orders in under 60 minutes. This speed is due primarily to large-scale, born-digital logistics integrators like Cainiao. A smart network, Cainiao leverages independently owned hardware and infrastructure such as warehouses and delivery vans and optimizes them across the country. These operations have several key features.

Gig workers. Last mile or local delivery to a customer's door, which is usually expensive, is a major challenge. In China this link in the supply chain is supplied by migrant workers (from the provinces) on bicycles or electric scooters. They are typically employed on a piecework basis by third-party logistics companies such as Meituan, which serves small grocers and restaurants, or by large online-off-line chains such as Hema Fresh. These workers constitute a large and cheap labor pool that few other countries can match.

Sophisticated routing technology. They may be riding bikes, but the workers are guided by routing software that speedily provides a large quantity of information. Once an order is placed, a middleware system subtracts it from the inventory, puts the information into the company's CRM system, chooses the nearest outlet for fast delivery, and alerts the nearest delivery worker. The delivery worker's app will even specify which stairs to take inside a building to get to the right floor, as instructions are passed on by the customer. As a result, delivery in China is extremely fast. Such efficiency enables workers to earn a good living, while employers can track each employee's contribution.

Free to the customer. The scale effects and highly efficient logistical operations keep delivery costs so low that most merchants can afford to cover them completely.



(Fresh-food delivery is an exception: It costs six times as much as regular delivery in China.) If a purchase exceeds a certain amount (typically \$15), or if a customer has brand membership through a loyalty program or for a token enrollment price, he or she can enjoy free delivery with speed.

Options for the West. Again, Western companies do not have access to an equivalent workforce, and navigating urban layouts can be challenging (imagine bike deliveries in Los Angeles). What's more, stricter labor regulation may make it harder to rely on gig workers. (That can be overcome, as evidenced by California's passage in November 2020 of Proposition 22, which preserved the nonemployee status of ride-share drivers.)

Companies will have to compensate by leaning more into tech. Many could digitize the back-end supply chain by deploying cloud-based warehouse management systems that are integrated across stores, enabling faster and more-flexible delivery services. Big data analytics will also contribute—subject to privacy constraints. In time, technology may even make it possible to take labor out of the equation, as autonomous vehicles and drones become feasible for

physical delivery. For example, Ericsson, Einride, and Telia have partnered to launch 5G-powered driverless and environmentally friendly trucks. Nuro, a self-driving delivery start-up, is teaming up with Domino's, Walmart, and CVS. And in the field of logistics competitors are working together to build huge data pools that will enable faster, better predictive analytics—as seen in the partnership of the rival vehicle telematics companies Geotab and Webfleet Solutions.

5 ALWAYS STAY CLOSE TO THE CUSTOMER

In China customer loyalty in digital retailing is generated in large part by extraordinarily high levels of after-sales engagement by companies and loyalty programs that are integrated into both e-commerce channels and social media. Companies also work with influencers and cultivate fan communities of their own.

Radical engagement. Chinese retailers truly treat the customer as royalty. Most online shops allow free returns with no explanation within seven days of a purchase (excluding fresh produce). Customers may choose a convenient



pickup time for their returned products. Retailers offer instantaneous personal assistance via chatbots. Free hotline support is readily available, often 24/7. For products that require on-site services (such as installation), customers may choose days and precise time slots. These services are often supplied by local shops that carry the brand. For example, the wooden-door producer Tata Pravesh treats its traditional distributors as after-sale service providers for customers purchasing from the online store. The distributors get the same financial return they would if the products had been sold in their off-line shops.

Integration across entry points. Customers have as many touchpoints with loyalty programs as they do with brands. Coupons and digital red envelopes (traditionally exchanged by relatives and friends during celebrations) are daily offerings on large platforms such as Taobao, Tmall, JD.com, and Meituan (for restaurant food order and delivery). Online shops offer deep discounts every day during certain hours to attract return customers and discounts for customers who make successful recommendations to

friends on social media. Loyalty programs are integrated across all digital channels.

Influencer relationships. As noted, the Chinese rely heavily on key opinion leaders' recommendations; in many cases KOLs command more loyalty than the brands they recommend. Chinese retailers piggyback on that relationship by cooperating with these KOLs, who often become associated with a particular brand. Some KOLs cut across product categories, representing brands for each. For example, Viya sells everything from makeup and shoes to rice and even houses. She sold goods worth \$4.5 billion in 2020 with a team of just 500 people.

Fan marketing. The home electronics company Xiaomi has created a huge base of loyal users who help promote the brand. Its fans will buy anything Xiaomi sells. Now the company has its own online ecosystem, having partnered with more than 300 producers to sell their products to its fans under the Xiaomi brand. On April 1, 2021, Xiaomi announced that it would invest \$10 billion over the next 10 years to produce smart electric cars. The company has sharpened its



industrial and project design capabilities to give products under its brand name a similar look and feel at an affordable price. This type of fan economy is steadily growing in China.

Options for the West. Western companies should leverage digital platforms to offer loyalty rewards more often, more explicitly, and in more places. They can also use their various channel partners to improve delivery and service. And continuous tech-enabled tracking of service delivery processes and customer engagement will motivate sales agents and other employees to attend to customers' needs.

In other aspects of customer engagement, Western retailers are not so far behind. The U.S. exercise equipment company Peloton's creation of an online community of users has been a major driver of its success and a source of innovative ideas. And at least one fast-fashion retailer, Shein, based in the United States but founded in China in 2008, uses influencers as its main channel to customers. It is hugely successful with this approach, generating multibillion-dollar revenues and challenging even Zara, the longtime leader in the sector.

WESTERN RETAILERS LAG their Chinese counterparts in leveraging customer data to make better business decisions, increase operational efficiency, and reduce costs. They need to integrate that data with off-line businesses so that customers are visible, identifiable, and traceable both online and off-line. Retailers need to establish contact with customers online through multiple touchpoints, including social media ecosystems, to increase their stickiness, loyalty, and activity. Digital retailing is an organizational transformation in the making. A famous Chinese saying has it that "a journey of a thousand miles begins with a single step." The new customer journey begins with many steps. 📍

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